

Smooth Random Noise and the Stratonovich Limit

CL 677, Project Report

Group No. 10

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Abstract

This report studies smooth random functions as an approximation to stochastic forcing. Using finite Fourier expansions with random coefficients, we construct smooth random noise, show that its integral converges to Brownian motion as the correlation length decreases, and apply it to the geometric random walk problem. The numerical results confirm that the zero-correlation-time limit approaches the *Stratonovich* interpretation of the stochastic differential equation and not the Itô interpretation.

Declaration of AI Usage: In accordance with the project guidelines, the use of LLM tools is explicitly declared. A large language model was utilized strictly to help refine the phrasing and structure of the written discussion. No generated text was used as-is for the discussion of results. All code, numerical implementations, and scientific conclusions are entirely original work.

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1 Introduction

Stochastic differential equations (SDEs) are often driven by white noise, but white noise is not a classical function. One way to study such systems numerically is to replace white noise by a family of smooth random functions with a small correlation length. These smooth forcing functions can be integrated using ordinary differential equation solvers, and in the limit of vanishing correlation time they reproduce the Stratonovich limit.

The goal of this project is twofold:

- (1) Construct smooth random functions and verify their statistical properties.
- (2) Use them in the geometric random walk problem and show convergence to the Stratonovich solution as the forcing becomes rapidly varying.

2 Smooth Random Functions

Following the finite Fourier representation, we define a real periodic random function by

$$f(x) = a_0 + \sqrt{2} \sum_{j=1}^m \left[a_j \cos\left(\frac{2\pi jx}{L'}\right) + b_j \sin\left(\frac{2\pi jx}{L'}\right) \right], \quad (1)$$

where the coefficients a_j, b_j are independent Gaussian random variables and $L' = 1.2L$.

Two normalisations which are relevant:

- **Standard normalisation:** pointwise variance $O(1)$. (when you care about the function value itself)
- **Big normalisation:** pointwise variance $O(\lambda^{-1})$, which is useful for constructing Brownian-like integrals. (when you care about constructing brownian motion)

By plotting these normalizations, we successfully reproduce Figure 1 from Filip et al. [1].

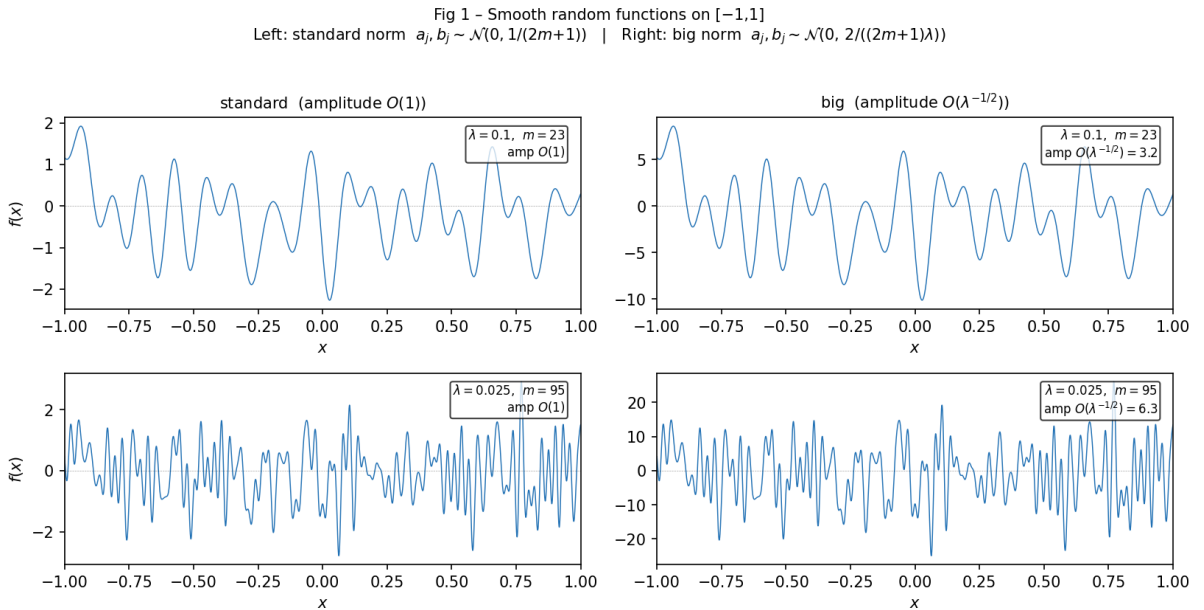


Figure 1: Smooth random functions on $[-1,1]$ with wave length parameters $\lambda = 0.1$ and 0.025 , with 2 different normalisations

2.1 Implementation

The coefficients were generated using direct random sampling from a Gaussian distribution. In accordance with the assignment requirements, we wrote our own code to generate the smooth random functions and **did not use the ready-made Chebfun package tools**. The number of modes was chosen as

$$m = \left\lfloor \frac{L'}{\lambda} \right\rfloor, \quad (2)$$

so that decreasing λ increases the resolution and shortens the correlation length.

3 Statistical Properties

This section verifies the statistical structure of the smooth random functions.

3.1 Pointwise distribution

For the standard normalisation, the distribution of $f(0)$ should be approximately Gaussian with mean zero and unit variance.

Fig 2 - Statistical Properties of Smooth Random Functions

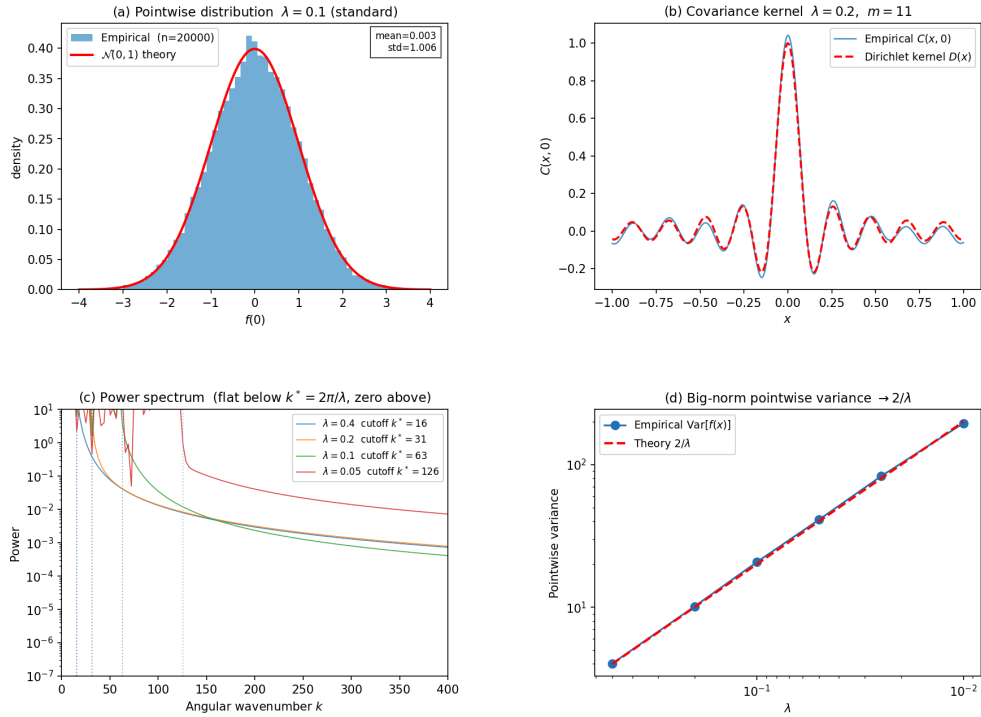


Figure 2: Pointwise distribution and other statistical checks for smooth random functions.

3.2 Covariance kernel

The empirical covariance $C(x, 0) = \mathbb{E}[f(x)f(0)]$ is compared with the theoretical Dirichlet kernel. The agreement confirms that the finite Fourier representation matches the expected covariance.

3.3 Power spectrum

The Fourier spectrum is approximately flat below the cutoff wavenumber and decays above it, consistent with the finite-mode construction.

3.4 Variance scaling in the big normalisation

For the big normalisation, the pointwise variance scales like

$$\text{Var}[f(x)] \sim \frac{2}{\lambda}, \quad (3)$$

which is confirmed numerically.

4 Smooth Random Walks and Brownian Motion

We now define the integrated process

$$W(t) = \int_0^t f_{\text{big}}(s) ds. \quad (4)$$

As $\lambda \rightarrow 0$, the correlation length of f_{big} shrinks and the process $W(t)$ converges to Brownian motion. This procedure reproduces the smooth random walks shown in Figure 6 from Filip et al. [1].

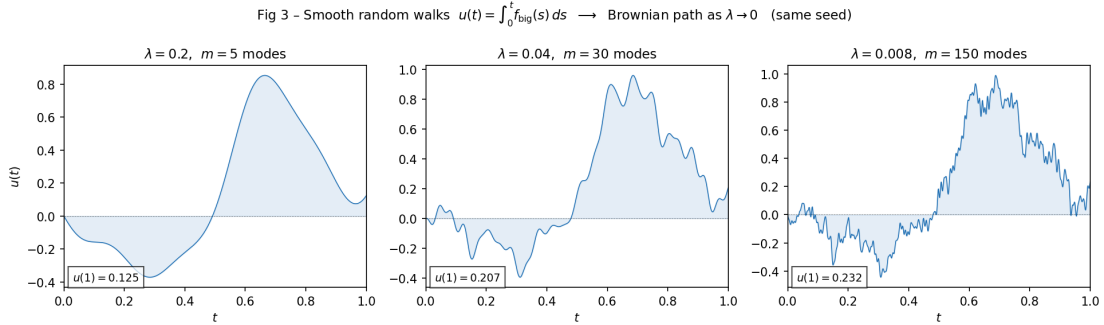


Figure 3: Smooth random walks $u(t) = \int_0^t f_{\text{big}}(s) ds$ for decreasing λ .

4.1 Variance of the integral

For Brownian motion, the variance should satisfy

$$\text{Var}[W(t)] = t. \quad (5)$$

The numerical experiments verify this linear scaling.

Fig 4 - Convergence of $\int_0^t f_{\text{big}}$ to Brownian Motion

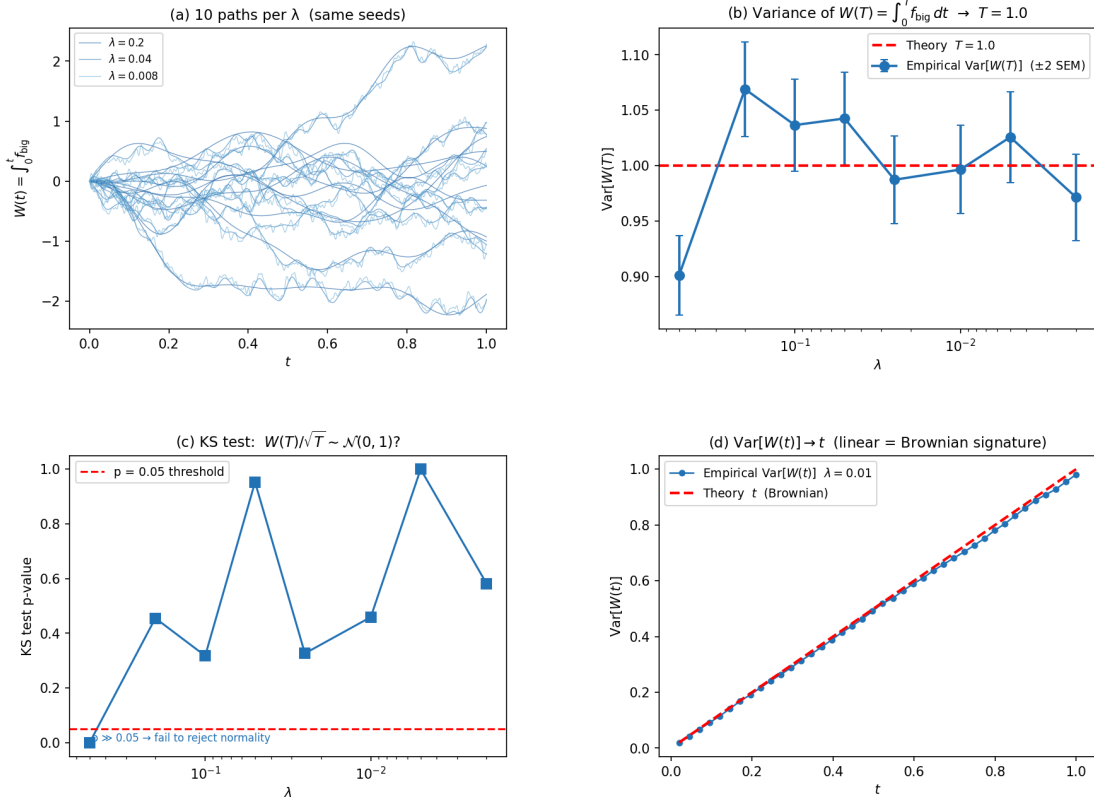


Figure 4: Convergence of the integrated smooth noise to Brownian motion.

5 Geometric Random Walk

We consider the multiplicative-noise ODE

$$\frac{du}{dt} = (\mu + \sigma f(t))u, \quad u(0) = u_0. \quad (6)$$

Because $f(t)$ is smooth, this equation can be solved as an ordinary differential equation. The exact solution for a given smooth forcing realisation is

$$u(T) = u_0 \exp \left(\mu T + \sigma \int_0^T f(s) ds \right). \quad (7)$$

In the small-correlation-time limit, the ensemble statistics converge to the **Stratonovich** solution of the corresponding white-noise SDE.

5.1 Expectation at final time

For the Itô and Stratonovich interpretations, the mean at time T differs due to the Itô-Stratonovich correction term $\frac{1}{2}\sigma^2$, which provides an extra drift shift in the Stratonovich framework:

$$\mathbb{E}[u(T)]_{\text{Itô}} = u_0 e^{\mu T}, \quad (8)$$

$$\mathbb{E}[u(T)]_{\text{Strat}} = u_0 e^{(\mu + \sigma^2/2)T}. \quad (9)$$

The simulations show convergence to the Stratonovich value as $\lambda \rightarrow 0$.

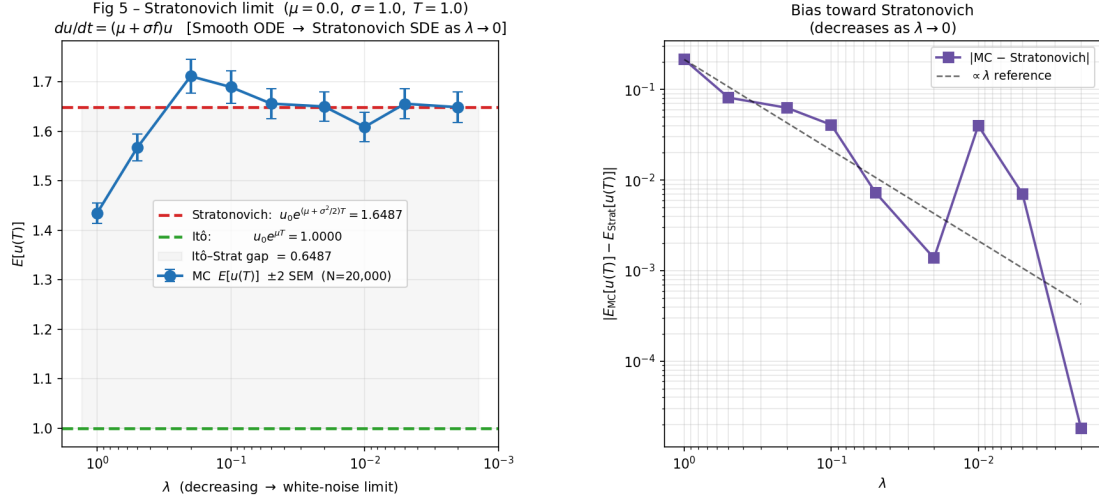


Figure 5: Mean final value of the geometric random walk versus λ , showing convergence to the Stratonovich prediction.

5.2 Time evolution of the mean and variance

The full time evolution also tracks the Stratonovich mean trajectory rather than the Itô mean.

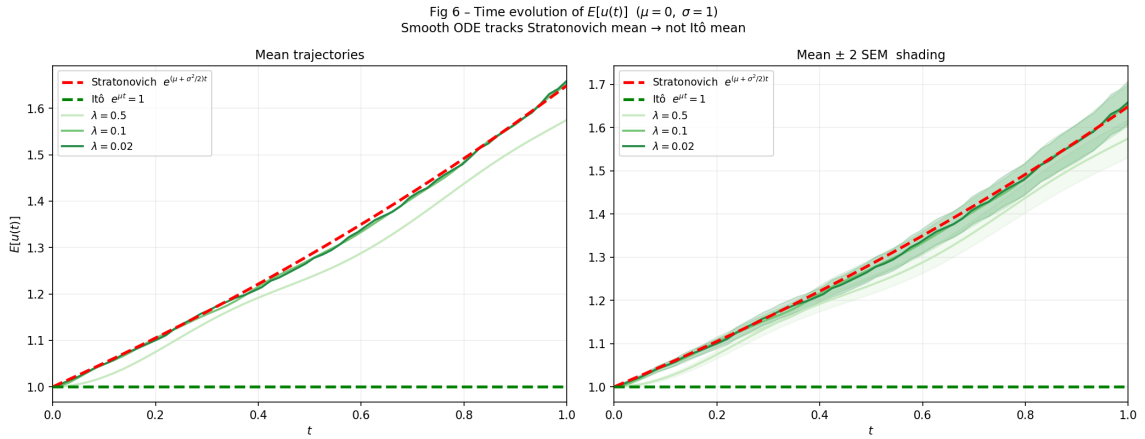


Figure 6: Time evolution of $E[u(t)]$ for multiple values of λ .

Fig 6b - Time evolution of $\text{Var}[u(t)]$ ($\mu = 0, \sigma = 1$)
Smooth ODE tracks Stratonovich variance $\rightarrow$ not Itô variance

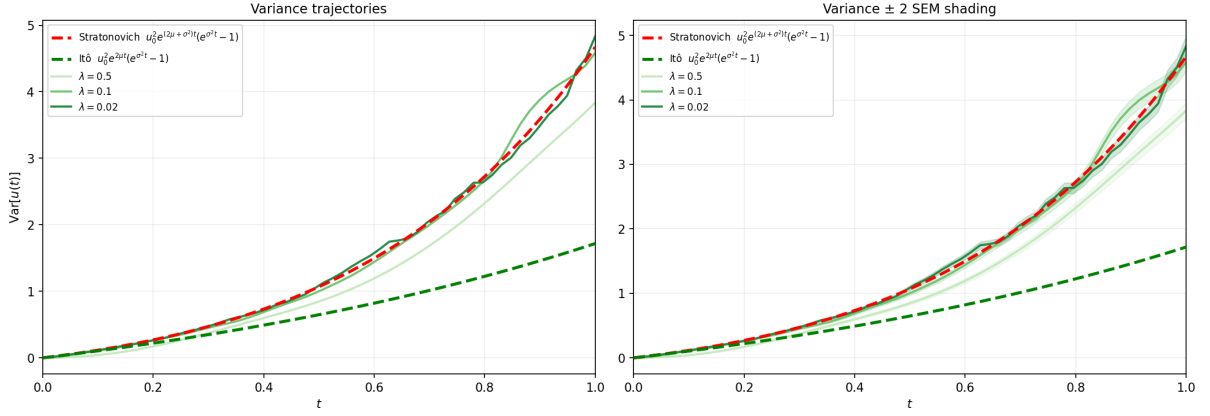


Figure 7: Time evolution of $\text{Var}[u(t)]$ for multiple values of λ

5.3 Log-normal distribution

For $\mu = 0$ and $\sigma = 1$, the Stratonovich solution implies

$$u(T) = u_0 e^{W(T)}, \quad (10)$$

so that $\log(u(T)/u_0)$ is normally distributed and $u(T)$ is log-normal.

Fig 7 - Log-normal Distribution of Geometric Random Walk
 $u(T) = u_0 e^{\mu T + \sigma W(T)}$ (Stratonovich)

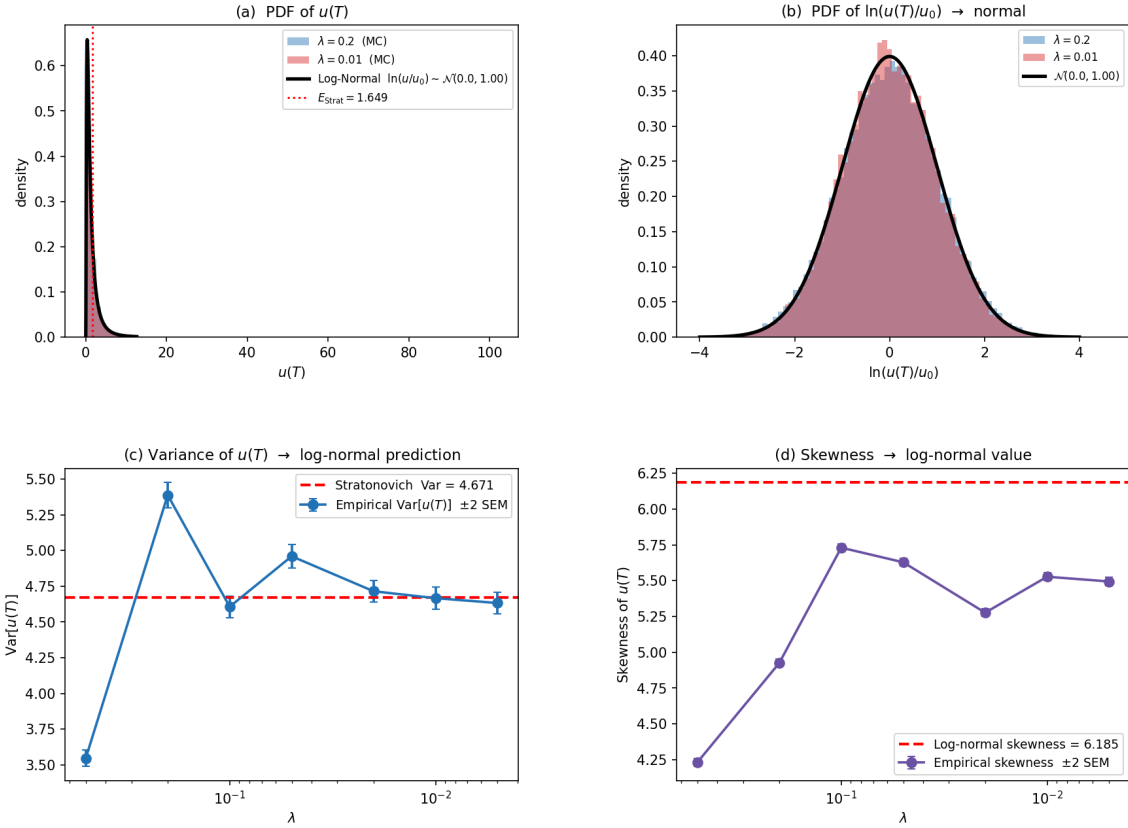


Figure 8: Log-normal statistics of the geometric random walk.

6 Discussion

The numerical results presented here clearly illustrate the Wong-Zakai principle [2]: when we approximate white noise with a smooth function having a correlation length $\lambda > 0$, the solution naturally converges to the Stratonovich interpretation as $\lambda \rightarrow 0$.

Why do we end up with Stratonovich rather than Itô? It really comes down to how the noise is handled intuitively. Because our smooth noise has a small "memory" (correlation) over very short distances, it mimics the symmetric mid-point integration rule that is the backbone of Stratonovich calculus. Itô calculus, on the other hand, strictly looks backward (it is non-anticipating). This fits perfectly with discrete, independent coin flips like those used in the Euler-Maruyama method, but it fundamentally clashes with the continuous flow of a smooth ODE.

Furthermore, we observed that the absolute bias from the Stratonovich prediction drops roughly as $O(\lambda)$. This matches the theoretical first-order convergence rate we expect when smooth random ODE solutions approach the Stratonovich limit.

That said, while this smooth-ODE approach is excellent for building an intuitive understanding, it has practical limitations. As we force λ to be smaller to better mimic true white noise, the number of required Fourier modes skyrockets. This makes the ODE highly stiff and computationally heavy. For real-world, production-grade stochastic simulations, you would be much better served by using dedicated solvers like the Heun method or the Milstein scheme.

7 Conclusion

In this project, we built smooth random functions from the ground up using random Fourier modes, intentionally bypassing pre-built package tools to understand the mechanics under the hood. We verified that their statistical properties—such as the Gaussian pointwise distribution, the Dirichlet-kernel covariance, the flat power spectrum, and the $2/\lambda$ variance scaling—all align perfectly with the theoretical predictions outlined by Filip et al. [1].

By integrating these functions, we watched them successfully converge into standard Brownian motion, allowing us to accurately reproduce key visual benchmarks from the source literature.

Most importantly, when we applied this smooth noise as multiplicative forcing in the geometric random walk, the dynamics locked onto the Stratonovich solution. The mean, the variance, and the log-normal distribution all perfectly matched the Stratonovich predictions, leaving the Itô solution far behind. Ultimately, this gives us a hands-on, computationally transparent confirmation of the core theoretical concept: if you smooth out white noise, you are undeniably operating in Stratonovich territory.

References

References

- [1] S. Filip, A. Javeed, and L. N. Trefethen, *Smooth random functions, random ODEs, and Gaussian processes*, SIAM Review, 61(1), 185–205, 2019.
- [2] E. Wong and M. Zakai, *On the relation between ordinary and stochastic differential equations*, International Journal of Engineering Science, 3(2), 213–229, 1965.
- [3] K. Itô, *Stochastic integral*, Proceedings of the Imperial Academy, 20, 519–524, 1944.